OnePrompt-All: One Click to All Same-Concept Instances

A training-free point-conditioned grounding framework for SAM 3

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Problem

Research question

Can one point trigger multi-instance segmentation in SAM 3?

OnePrompt-All asks whether a single point can trigger multi-instance grounding. The click anchors SAM 3's pretrained visual representation and elicits a concept-level response without category text or training.

Interaction paradigm

PVS
Point → one instance

PCS
Text → concept set

OnePrompt-All
Point → multi-instance response

Method

OnePrompt-All uses a positive click as a *geometric condition* rather than a category label. The click anchors the target in SAM 3's pretrained visual representation, while learned object queries search the full image in parallel for candidate instance masks. Lightweight post-processing then produces the final instance set without additional training.

Point-conditioned grounding

The click supplies sparse spatial evidence about the intended object. SAM 3 injects this cue into the grounding pathway, so the detector is conditioned by the selected location while still searching globally.

Candidate generation

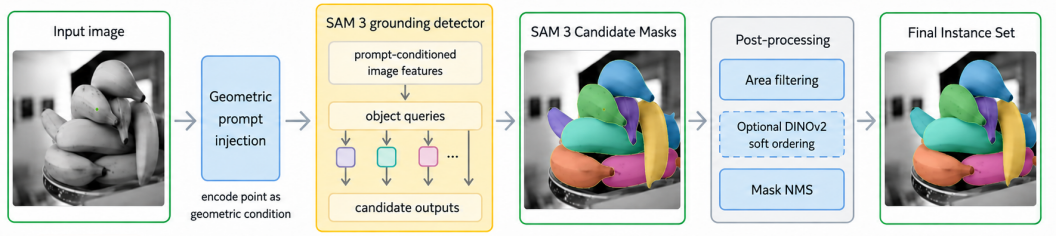
Multiple object queries attend to conditioned image features in parallel. Each produces a scored candidate mask, yielding valid instances, weak fragments, and duplicate responses.

Filtering and deduplication

Area filtering first removes trivial fragments using a class-agnostic size rule. Mask NMS then suppresses highly overlapping predictions and keeps the stronger candidate, yielding independent instance masks.

Optional DINOv2 ordering

Before NMS, DINOv2 may softly reorder candidates by similarity to the clicked seed. It is auxiliary ranking rather than a hard semantic filter, so the core pipeline remains training-free and category-agnostic.



Results

Evaluation

Integrated OnePrompt-All Benchmark: 2,220 images, 232 classes, and 11,100 trials. Each run uses RGB + one positive click; ground truth is used only for prompt sampling and scoring.

Main findings

OnePrompt-All achieves **F1 0.667** and **mIoU 0.601**, with balanced precision (0.687) and recall (0.712) across the integrated benchmark. Remaining errors are mainly caused by false candidates and sensitivity to click location.

Method	Benchmark	Images	Classes	Runs	F1	F1 SD	Prec.	Recall	mIoU	IoU SD
OnePrompt-All	Integrated Benchmark	2,220	232	11,100	**0.667**	0.280	0.687	0.712	**0.601**	0.279

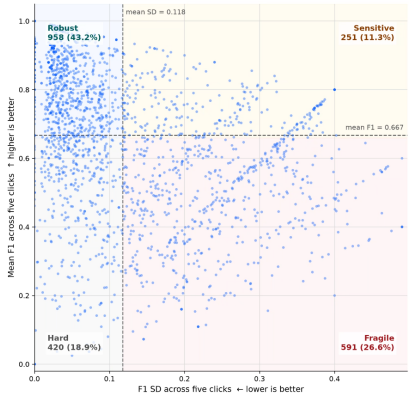


Fig. 2. Accuracy vs. click robustness across 2,220 images; dashed lines mark dataset means.

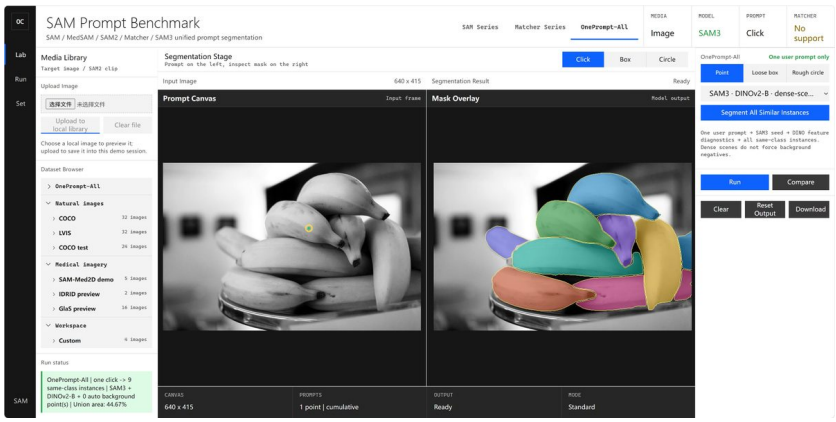


Fig. 3. Interactive workbench for prompting, model comparison, and mask inspection.

Conclusion & Discussion

Conclusion

A single point can trigger a multi-instance response in SAM 3, enabling **training-free multi-instance segmentation**.

Discussion

Global object queries recover related instances, while area filtering and mask NMS suppress fragments and duplicates.

Limitations

Performance still depends on click placement, occlusion, appearance variation, and distractors; candidate calibration remains future work.